Prayas JEE AIR (2025)

WEEKLY TEST 32

DURATION : 180 Minutes | **DATE : 18/01/2025** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (4)
2. (1)
3. (3)
4. (3)
5. (4)
6. (2)
7. (3)
8. (2)
9. (2)
10. (1)
11. (3)
12. (2)
13. (2)
14. (1)
15. (1)
16. (3)
17. (2)
18. (2)
19. (2)
20. (1)
21. (4)
22. (2)
23. (5)
24. (13)
25. (20)

CHEMISTRY

26. (3)
27. (2)
28. (3)
29. (2)
30. (1)
31. (2)
32. (4)
33. (2)
34. (2)
35. (2)
36. (1)
37. (2)
38. (3)
39. (3)
40. (1)
41. (2)
42. (2)
43. (2)
44. (1)
45. (1)
46. (3)
47. (7)
48. (3)
49. (24)
50. (2)

MATHEMATICS

51. (4)
52. (3)
53. (1)
54. (3)
55. (1)
56. (1)
57. (2)
58. (2)
59. (2)
60. (1)
61. (2)
62. (1)
63. (2)
64. (4)
65. (3)
66. (1)
67. (3)
68. (1)
69. (4)
70. (3)
71. (3)
72. (3)
73. (7)
74. (4)
75. (4)

SECTION-I (PHYSICS)

1. **(4)**

$S = t^3 + 5$

$\therefore$ Speed, $v = \frac{ds}{dt} = 3t^2$

and rate of change of speed $= \frac{dv}{dt} = 6t$

$\therefore$ tangential acceleration at $t = 2$ s,

$a^t = 6 \times 2 = 12$ m/s^2

At $t = 2$ s, v $= 3(2)^2 = 12$ m/s

$\therefore$ centripetal acceleration, $a_c = \frac{v^2}{R} = \frac{144}{20}$ m/s^2

$\therefore$ net acceleration $= \sqrt{a_t^2 + a_i^2}$

≈ 14 m/s^2

2. **(1)**

Truth table for given combination is

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	1

This comes out to be truth table of OR gate.

3. **(3)**

$T - mg\cos\theta = \frac{mu^2}{\ell}$

$u^2 = 2gl\cos\theta$

$T = 3mg\cos\theta$

4. **(3)**

$\Delta P = 10 - (10) \times 0.2$

$\Rightarrow$ $8 = 1 \times v$

$\Rightarrow$ $v = 8$ m/sec

5. **(4)**

Angular momentum about point of contact will be conserved

$L_i = L_f$

$$\left(\frac{2}{5}Mr^2\right)\frac{v_0}{2r} + Mv_0 r = \left(\frac{2}{5}Mr^2\right)\frac{v}{r} + Mrv$$

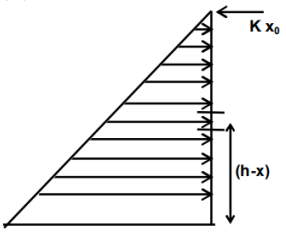
$$\Rightarrow v = \frac{6}{7}v_0$$

6. **(2)**

Escape velocity, $v = \sqrt{2gR}$

$$\therefore \quad \frac{v_E}{v_M} = \sqrt{\frac{2g_E R_E}{2g_M R_M}} = \sqrt{6\times 4} = \sqrt{24}$$

7. **(3)**



$$kx_0 h = \int_0^h (bdx)\rho g x(h-x)$$

$$\Rightarrow kx_0 h = b\rho g\int_0^h (hx - x^2)\,dx$$

$$= b\rho g\left[h\frac{h^2}{2} - \frac{h^3}{3}\right]$$

$$\Rightarrow kx_0 h = b\rho g\frac{h^3}{6} \Rightarrow x_0 = \frac{b\rho g h^2}{6k}$$

$$PE = \frac{1}{2}kx_0^2 = \frac{1}{2}k\frac{b^2\rho^2 g^2 h^4}{36k^2} = \frac{b^2\rho^2 g^2 h^4}{72k}$$

8. **(2)**

(i) $(\text{stress})_A = (\text{stress})_B$

$$\Rightarrow \frac{T_A}{a_A} = \frac{T_B}{a_B} \;\&\; T_A x_1 = T_B(\ell - x_1)$$

$$\Rightarrow x_1 = \frac{2\ell}{3}$$

(ii) $(\text{strain})_A = (\text{strain})_B$

$$\Rightarrow \frac{Y_A}{\left(\frac{T_A}{a_A}\right)} = \frac{Y_B}{\left(\frac{T_B}{a_B}\right)} \;\&\; T_A x_2 = T_B(\ell - x_2)$$

$$\Rightarrow x_2 = \frac{6\ell}{7}$$

Hence $x_1 : x_2 = 7 : 9$

$a \rightarrow$ Area of cross - section
$T \rightarrow$ Tension in the wire
$Y \rightarrow$ Youngs Modulus

9. **(2)**

We know that $\frac{dQ}{dt} = KA\frac{d\theta}{dx}$

In steady state flow of heat

$$d\theta = \frac{dQ}{dt}\cdot\frac{1}{kA}dx$$

$\Rightarrow \theta_H - \theta = k'x \Rightarrow \theta = \theta_H - k'x$

Equation $\theta = \theta_H - k'x$ represents a straight line.

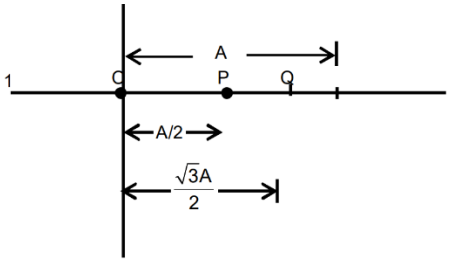
10. **(1)**

$Q = Q_1 + Q_2$

$$\frac{n_1 + n_2}{\gamma_m - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

$$\gamma_m = \frac{3}{2}$$

11. **(3)**



$$\frac{A}{2} = A\sin\omega t_1 \Rightarrow t_1 = \frac{\pi}{6\omega}$$

$$\frac{\sqrt{3}}{2}A = A\sin\omega t \Rightarrow t_2 = \frac{\pi}{3\omega}$$

$$t_2 - t_1 = \frac{\pi}{6\omega} = \frac{\pi \times T}{6 \times 2\pi} = T/12$$

12. **(2)**

Binding energy = $(M_O - 8M_P - 9M_N)\,C^2$

13. **(2)**

Apply shell theorem the total charge upto distance r can e calculated as followed

$$dq = 4\pi r^2 \cdot dr \cdot \rho = 4\pi r^2 \cdot dr \cdot \rho_0\left[\frac{5}{4} - \frac{r}{R}\right]$$

$$= 4\pi\rho_0\left[\frac{5}{4}r^2 dr - \frac{r^3}{R}dr\right]$$

$$\int dq = q = 4\pi\rho_0\int_0^r\left(\frac{5}{4}r^2 dr - \frac{r^3}{R}dr\right)$$

$$= 4\pi\rho_0\left[\frac{5}{4}\frac{r^2}{3} - \frac{1}{R}\frac{r^4}{4}\right]$$

$$E = \frac{kq}{r^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\cdot 4\pi\rho_0\left[\frac{5}{4}\frac{r^2}{3} - \frac{1}{R}\frac{r^4}{4}\right]$$

$$E = \frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3} - \frac{r}{R}\right)$$

14. **(1)**

All the capacitors are in parallel, then

Ceq = $C_1 + C_2 + C_3$

Ceq = 10 μF

15. **(1)**

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta l}{l}$$

$$\Rightarrow \frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta l}{l} \times 100 = 3\%$$

16. **(3)**

When bar becomes vertical

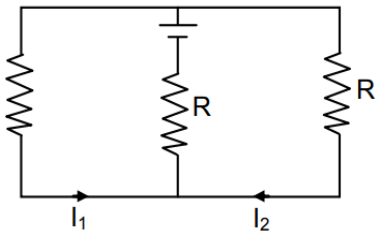
$$\omega = \sqrt{\frac{3g}{\ell}}$$

So $V_q = \sqrt{3g\ell}$ and $V_c = \sqrt{\frac{3gl}{4}}$

$$F - mg - qV\,B = \frac{mV_c^2}{\frac{l}{2}}$$

$\Rightarrow F - 10 - 3 = 15$

17. **(2)**



A moving conductor is equivalent to a battery of emf = $vB\ell$ (motion emf)

Equivalent circuit

$I = I_1 + I_2$

Applying Kirchoff's law

$I_1R + IR - vB\ell = 0$ …(1)

$I_2R + IR - vB\ell = 0$ …(1)

Adding (1) and (2)

$2IR + IR = 2vB\ell$

$$I = \frac{2vB\ell}{3R}; I_1 = I_2 = \frac{vB\ell}{3R}$$

18. **(2)**

$P_{avg} = V_{rma}\, I_{rma} \cos(\phi)$

$$P_{avg} = \frac{V_{rms}}{z^2}R = \frac{V^2R}{(R^2 + (\omega l)^2)}$$

19. **(2)**

Average energy density is given by,

$$U = \frac{1}{2}\int_0 E^2 = \frac{1}{2}\int_0\left(\frac{E_0}{\sqrt{2}}\right)^2 = \frac{1}{4}\in E^2$$

$$U = \frac{1}{4} \times 8.85 \times 10^{-12} \times 1^2$$

$$U = 2.2 \times 10^{-12}\ \text{J/m}^3$$

20. **(1)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$$

$$\frac{dv}{dt} = -\frac{v^2}{u^2}\left(\frac{du}{dt}\right)$$

$f = 20$ cm

$$\frac{1}{u} + \frac{1}{-280} = \frac{1}{20}$$

$$v_1 = -\left(\frac{280}{15 \times 280}\right)^2 \times 15 = \frac{1}{15}\text{m/s}$$

21. **(4)**

$$\frac{I}{I_{max}} = \cos^2\left(\frac{\phi}{2}\right) = \frac{3}{4}$$

$$\Rightarrow \frac{I}{I_0} = \frac{3}{4}$$

$$\Rightarrow \frac{3I_0}{I} = 4$$

22. **(2)**

$$\frac{1}{\lambda_0} = k\left(\frac{5}{36}\right)$$

$$\frac{1}{\lambda^1} = k\left(\frac{3}{16}\right)$$

$$\frac{\lambda^1}{\lambda_0} = \frac{20}{27}$$

23. **(5)**

The amplitude of the resultant wave is

$\sqrt{3^2 + 4^2 + 2\times3\times4\times\cos 90}$

= 5 m

24. **(13)**

$$\Rightarrow \frac{12400}{4500} - \phi = 13.6\left(\frac{5}{36}\right)$$

$$\Rightarrow \phi = \frac{13}{15}\text{eV}.$$

25. **(20)**

After the upper spring is cut, the block 1 and 2 will move with same acceleration.

$\therefore \quad a = 2g$

SECTION-II (CHEMISTRY)

26. **(3)**

When HCl reacts with finely powdered iron, it forms ferrous chloride not ferric chloride

27. **(2)**

$HgCl_2$ will not give chromyl

28. **(3)**

H_2O is weak field ligand while CN^- is a strong field ligand. In complex (i), the distribution is $t_{2g}^3 e_g^2$, i.e. all electrons are unpaired while in (ii) complex, the distribution is $t_{2g}^5 e_g^0$, i.e. two t_{2g} orbitals are paired while one is unpaired.

29. **(2)**

Non ideal solution with negative deviation from Raoult's law

30. **(1)**

$3A + B \rightarrow 2C$

$$r = \frac{1}{3}\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{1}{2}\frac{d[C]}{dt} = K'[A][B]$$

$$-\frac{d[A]}{dt} = 3K'[A][B] = K[A][B]$$

i.e. $K = 3K'$

So that $\frac{d[C]}{dt} = 2K'[A][B] = \frac{2}{3}K[A][B]$

31. **(2)**

$$\Lambda^\circ_{BaSO_4} = \kappa \times \frac{1000}{\text{Conc(normality)}}$$

$$\text{Normality} = \frac{\kappa\times1000}{\Lambda^\circ_{BaSO_4}} = \frac{1000\times8\times10^{-5}}{400} = 2\times10^{-4}$$

$$\text{Molarity} = \frac{\text{Normality}}{2} = 10^{-4}\text{M (Solubility)}$$

$$\text{Ksp} = S^2 = 10^{-8}\text{M}^2$$

32. **(4)**

Cyclic amines are more basic than non cyclic amines

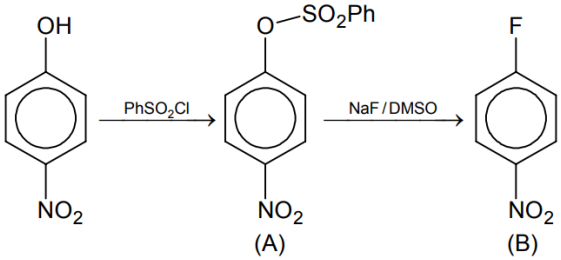
33. **(2)**

It does not have α-hydrogen atom connected to sp^3 carbon.

34. **(2)**

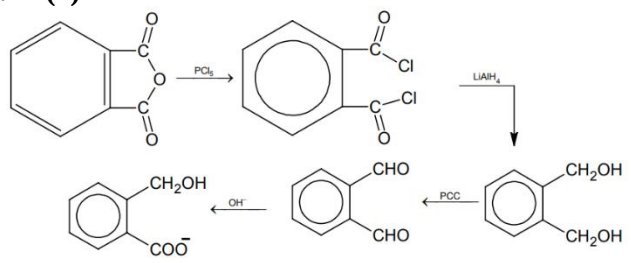
NH_2^- behaves as a base and E_2 elimination takes place.

35. **(2)**



F^- is very reactive unsolvated which can displace $PhSO_3^-$ which being a good leaving group in presence of strong withdrawing group $-NO_2$.

36. **(1)**



37. **(2)**

II > I > IV > III

38. **(3)**

α-D glucose ⇌ β-D Glucose

1–x x

$$K = \frac{x}{1-x} = 1.8 \Rightarrow x = 0.643$$

Now $(1 - 0.643) \times 112 + 0.643 \times \theta = 59$

$0.643 \times \theta = 59 - 42.84$

$$\theta = \frac{16.16}{0.643} = 29.57$$

39. **(3)**

$$2Ag_2CO_3 \xrightarrow{\Delta} 4Ag + 2CO_2 + O_2$$

552 gm Ag_2CO_3 → 432 g of Ag

$$\therefore \quad 2.76 \text{ g } Ag_2CO_3 \rightarrow \frac{432}{552} \times 2.76 = 2.16 \text{ g}$$

40. **(1)**

(A) → (P); (B) → (Q); (C) → (R); (D) → (S)

41. **(2)**

Refer theory

42. **(2)**

CS_2 is linear

43. **(2)**

M	+	O_2	→	Metallic Oxide (O^{2-})
80 gm		20 gm		100 gm

No. of equivalents of M = no. of equivalents of O_2

$$\Rightarrow \frac{80}{E} = \frac{20}{8}$$

$$\therefore E = \frac{80 \times 8}{20} = 32$$

44. **(1)**

$\Delta H > 0, \Delta S > 0$

⇒ Reaction may be non-spontaneous at 25°C

$\Delta G = \Delta H - T\Delta S$

$= 180 - 298 \times 150 \times 10^{-3}$

$= 135.3 > 0$

= Non-spontaneous

To make it spontaneous ($\Delta G < 0$). We have to increase the temperature.

$$T = \frac{\Delta H}{\Delta S} = \frac{180 \times 10^3}{150}$$

= 1200 K = 927°C

45. **(1)**

(A) → (R); (B) → (P); (C) → (Q); (D) → (S)

46. **(3)**

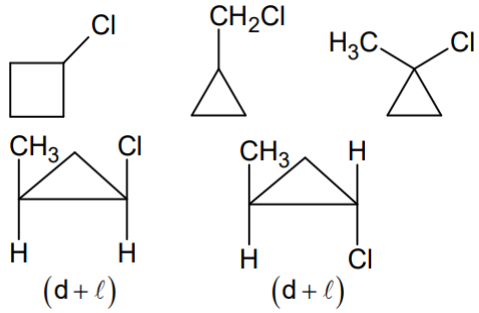
$\Delta T_b = Kb \times m \times i$

$2.08 = 0.52 \times 1 \times i$

$i = 4$

$K_3[Fe'''(CN)_6]$

47. **(7)**



Total number of isomers = 7.

48. **(3)**

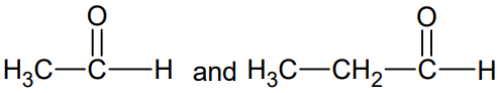
N_2O, NO and CO are neutral oxides remaining are acidic oxides.

49. **(24)**

$2C + H_2(g) \rightarrow C_2H_2(g)$

$\Delta H = 310 - [(2 \times 94) + 98] = 24$ kcal

50. **(2)**



react with Fehling solution.

SECTION-III (MATHEMATICS)

51. **(4)**

Centre and vertices only

52. **(3)**

$$\vec{r} \times \vec{b} - \vec{c} \times \vec{b} = 0$$

$$(\vec{r} - \vec{c}) \times \vec{b} = 0$$

$$\vec{r} - \vec{c} = \lambda \vec{b}$$

$$\vec{r} = \vec{c} + \lambda \vec{d}$$

$$\vec{r} = (1 + 3\lambda)\hat{i} + (1 - \lambda)\hat{j} + (3 + \lambda)\hat{k}$$

$$\vec{r} \cdot \vec{a} = 0$$

$\Rightarrow 2 + 6\lambda + 3 - 3\lambda - 3 - \lambda = 0$

$2\lambda = -2 \Rightarrow \lambda = -1$

$$\Rightarrow \vec{r} = -2\hat{i} + 2\hat{j} + 2\hat{k}$$

53. **(1)**

$$\left(x^{1/3}+1-\frac{x^{\frac{1}{2}}+1}{x^{\frac{1}{2}}}\right)^{10}$$

$$\left(x^{\frac{1}{3}}-x^{\frac{-1}{2}}\right)^{10}$$

$${}^{10}C_r\ x^{\frac{10-r}{3}}(-1)^r\ x^{\frac{-r}{2}}$$

$20 - 2r - 3r = 0$

$r = 4$

$$\therefore\ {}^{10}C_4=\frac{10\times9\times8\times7}{24}=210$$

54. **(3)**

$3x^3 + bx^2 + bx + 3 = 0$

$\Rightarrow 3(x^3 + 1) + bx(x + 1) = 0$ has roots $-1, 2, \frac{1}{2}$

$$\text{So, } \lim_{x\to\frac{1}{2}}\frac{e^{f(x)}-1}{2x-1}=\lim_{x\to\frac{1}{2}}\frac{3x^3+bx^2+bx+3}{2x-1}$$

$$=\lim_{x\to\frac{1}{2}}\frac{3(x+1)(x-2)\left(x-\frac{1}{2}\right)}{2\left(x-\frac{1}{2}\right)}$$

$$=\frac{3}{2}\times\frac{3}{2}\times\frac{-3}{2}=\frac{-27}{8}$$

55. **(1)**

Reflexive and symmetric

56. **(1)**

$x > -1 \Rightarrow 1 + x > 0$

$\Rightarrow (1 + f(x)) f''(x) > 0$

$\Rightarrow 1 + f(x)$ and $f''(x)$ are either both positive or both negative for $x > -1$

But $1 + f(0) = 2 > 0$

$\Rightarrow f''(x) > 0$

$\Rightarrow f'(x)$ is increasing

$\Rightarrow f'(x) > f'(0) = 0$

57. **(2)**

We have $f(x) = (a - x^n)^{1/n}$ so that

$$f_0 f(x)=f((a-x^n)^{1/n})$$

$$=(a-((a-x^n)^{1/n})^n)^{1/n}=(a-(a-x^n))^{1/n}$$

$$\Rightarrow g(x)=\underbrace{f_0\ldots f(x)}_{2m\text{ times}}=x$$

$$\therefore\int\frac{dx}{(g(x))^n(1+(g(x))^n)^{1/n}}=\int\frac{dx}{x^n(1+x^n)^{1/n}}$$

$$=\int\frac{dx}{x^{n+1}(1+x^{-n})^{1/n}}=-\int\frac{dt}{nt^{1/n}}(u\sin gt=1+x^{-n})$$

$$=-\frac{1}{n}\left(\frac{t^{-1/n+1}}{-\frac{1}{n}+1}\right)+c=\frac{1}{1-n}(1+x^{-n})^{1-\frac{1}{n}}+c$$

58. **(2)**

Given $y = f(x)$, $z = g(x)$ we have

$$\frac{dy}{dz}=\frac{dy/dx}{dz/dx}=\frac{f'(x)}{g'(x)}\text{ and}$$

$$\frac{d^2y}{dz^2}=\frac{d}{dx}\left(\frac{f'}{g'}\right)\frac{dx}{dz}=\frac{g'f''-f'g''}{(g')^2}\cdot\frac{1}{g'}$$

59. **(2)**

$3x - 4y + 5 = 0$

$$3x-4y-\frac{9}{2}=0$$

$$\text{Distance}=\frac{5+\frac{9}{2}}{5}=1+0.9=1.9$$

Radeons = 0.95

60. **(1)**

$(x - 0)^2 + (y - k)^2 = k^2$

$\Rightarrow x^2 + (y - k)^2 = k^2$

$$2x+2(y-k)\frac{dy}{dx}=0$$

$$\Rightarrow\frac{dy}{dx}=-\frac{x}{y-k}$$

$$y-k=-\frac{xdx}{dy}$$

$$k=y+\frac{xdx}{dy}$$

$$\Rightarrow x^2+\left(y-\left(y+\frac{xdx}{dy}\right)\right)^2=\left(y+\frac{xdx}{dy}\right)^2$$

$$\Rightarrow x^2+x^2\left(\frac{dx}{dy}\right)^2=y^2+x^2\left(\frac{dx}{dy}\right)^2+\frac{2xydx}{dy}$$

$$x^2=y^2+\frac{2xydx}{dy}$$

$$(x^2-y^2)\frac{dy}{dx}-2xy=0$$

61. **(2)**

Given $\alpha < \beta < \gamma < \delta$ also $\sin\alpha = \sin\beta = \sin\gamma = \sin\delta = k$ and $\alpha, \beta, \gamma, \delta$ are smallest positive angles

$\therefore\quad \beta = \pi - \alpha,\ \gamma = 2\pi + \alpha,\ \delta = 3\pi - \alpha$

as $\sin\beta = \sin\alpha$ and $\beta > \alpha$

$\sin\beta = \sin\gamma$ and $\gamma > \beta$

$\sin\gamma = \sin\delta$ and $\delta > \gamma$

Putting these values in the given expansion, we have given expression

$$= 2\left(\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right) = 2\sqrt{1+\sin\alpha} = 2\sqrt{1+k}$$

62. (1)

$$x^3 - x < -a^2 + a - \frac{2}{3\sqrt{3}}$$

$$f(x) = x^3 - x$$

$$f'(x) = 3x^2 - 1 = 0$$

$$x = \pm\frac{1}{\sqrt{3}}$$

In positive region minimum value of

$$f(x) = \frac{1}{3\sqrt{3}} - \frac{1}{\sqrt{3}} = -\frac{2}{3\sqrt{3}}$$

So, $-a^2 + a - \frac{2}{3\sqrt{3}} > -\frac{2}{3\sqrt{3}}$

$a^2 - a < 0$

$a \in (0, 1)$

63. (2)

$$\underset{v}{\sqrt{3x-2}} + \underset{u}{\sqrt{4x+4}} = \underset{p}{\sqrt{5x+1}} + \underset{q}{\sqrt{4x-5}} \quad \ldots(1)$$

$u^2 - v^2 = x + 6 = p^2 - q^2$

$u - v = p - q \quad \ldots(2)$

solving (1) and (2)

we get, $2\sqrt{4x+4} = 2\sqrt{5x+1}$

$\Rightarrow \quad x = 3$

64. (4)

$$\lim_{x\to\infty} \frac{x^2 + 2x - 1 - x^2}{\sqrt{x^2+2x-1} + x}$$

$$= \lim_{x\to\infty} \frac{2x-1}{x\left(\sqrt{1+\frac{2}{x}-\frac{1}{x^2}} + 1\right)}$$

$$= \frac{2}{1+1} = 1$$

65. (3)

$0 \le \arg z \le \frac{\pi}{4}$, represent the region of complex plane lying in the first quadrant and bounded by x axis and the line $y = x$

$|2z - 4i| = 2|z - 2i|$

least value of $|z - 2i|$ is length of perpendicular from (0, 2) to $y = x$, which is $\sqrt{2}$

So the least value of $\sqrt{2}\,|\,2z - 4i\,|$ is 4

66. (1)

Let $f(x) = 2x^3 + 3x^2 - 12x + 3$

$\Rightarrow \quad f(x) = 0$ has three real roots (α, β, γ)

$$\alpha + \beta + \gamma = -\frac{3}{2}$$

$$\text{Centroid} = \left(\frac{\sum\alpha}{3}, \frac{\sum\alpha}{3}, \frac{\sum\alpha}{3}\right)$$

$$= \left(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right)$$

which lies on $x = y = z$.

67. (3)

If $a > b > c$ and $a^2 = b = \sqrt{c}$ then $b < 1$

$\Rightarrow \quad \cot^{-1} x < 1 \Rightarrow x > \cot 1$

68. (1)

$n(S) = {}^{64}C_3$

Let 'E' be the event of selecting 3 squares which from the letter 'L'

No ways of selecting squares consisting of 4 unit squares $= {}^7C_1 \times {}^7C_1 = 49$

Each square with 4 unit squares forms 4L – shapes consisting of 3 unit squares

$\therefore \quad n(E) = 7 \times 7 \times 4 = 196$

$$\therefore\ p(E) = \frac{196}{{}^{64}C_3}$$

69. (4)

$$x^2 + 9y^2 + 25z^2 = 15yz + 5zx + 3xy$$

$$(x)^2 + (3y)^2 + (5z)^2 - (x)(3y) - (3y)(5z) - (x)(5z) = 0$$

$$\frac{1}{2}[2(x)^2 + 2(3y)^2 + 2(5z)^2 - 2(x)(3y) - 2(3y)(5z) - 2(x)(5z)]$$

$$= \frac{1}{2}\left[(x-3y)^2 + (3y-5z)^2 + (x-5z)^2\right] = 0$$

$x - 3y = 0, 3y - 5z = 0,\ x - 5z = 0$

$$\frac{1}{x} = \frac{1}{3y} \text{ and } \frac{5}{3y} = \frac{1}{z} \text{ and } \frac{1}{5z} = \frac{1}{x}$$

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{3y} + \frac{5}{3y} = \frac{2}{y}$$

$\therefore \quad \frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

70. (3)

When $z = n + 1$

we can choose x, y from $\{1, 2, \ldots, n\}$

$\therefore$ when $z = n + 1$, x, y can be chosen in n^2 ways and $z = n$, x, y can be chosen in $(n-1)^2$ ways and so on

$$\therefore \quad n^2 + (n-1)^2 + \ldots + 1^2 = \frac{1}{6}n(n+1)(2n+1)$$

ways of choosing triplets

Alternatively triplets with $x = y < z$, $x < y < z$, $y < x < z$ can be chosen in

${}^{n+1}C_2$, ${}^{n+1}C_3$, ${}^{n+1}C_3$ ways.

There are

$${}^{n+1}C_2 + 2({}^{n+1}C_3) = {}^{n+2}C_2 + {}^{n+1}C_3$$

$$= 2({}^{n+2}C_3) - {}^{n+1}C_2.$$

71. (3)

Putting $x = 2$, $\frac{1}{2}$ and -1 successively

$$f(2) + f\left(\frac{1}{2}\right) = 3 \qquad \ldots(1)$$

$$f\left(\frac{1}{2}\right) + f(-1) = \frac{3}{2} \qquad \ldots(2)$$

and $f(-1) + f(2) = 0$

Solving, we get $f(2) = \frac{3}{4}$

72. (3)

On solving we have

$A = \cos^4\theta + 1 - \cos^2\theta + 1$

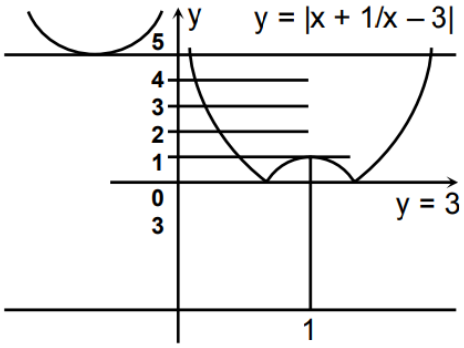
$$= \left(\cos^2\theta - \frac{1}{2}\right)^2 + \frac{7}{4} \Rightarrow A \in \left[\frac{7}{4}, 2\right]$$

73. (7)

$\Rightarrow$ $p - 3 = 0, 2, 3, 4$

$\Rightarrow$ $\Sigma p = 5 + 6 + 7 + 3 = 21$

$\Rightarrow$ $n = 7$



74. (4)

$$\left(x + \frac{1}{\alpha^2}\right)\left(x + \frac{1}{\beta^2}\right) + \frac{1}{\alpha\beta}$$

$$= x^2 + \left(\frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}\right)x + \frac{1}{(\alpha\beta)^2} + \frac{1}{\alpha\beta}$$

$$= x^2 + \left(\frac{t^4 + 4t}{4t^2}\right)x + \frac{1}{4t^2} - \frac{1}{2t}$$

Let $I = \int_{-1}^{2} x^2 dx + \frac{3}{2}\left(\frac{t^2}{4} + \frac{1}{t}\right) + 3\left(\frac{1}{4t^2} - \frac{1}{2t}\right)$

$$\Rightarrow \frac{dl}{dt} = \frac{3}{2}\left(\frac{t}{2} - \frac{1}{t^2} - \frac{1}{t^3} + \frac{1}{t^2}\right) = 0$$

$$\Rightarrow t^4 = 2 \Rightarrow t = \pm(2)^{1/4}$$

$$\Rightarrow \text{Min } I(t) = 3 + \frac{3}{2}\left(\frac{\sqrt{2}}{4} + \frac{1}{2^{1/4}} + \frac{1}{2\sqrt{2}} - \frac{1}{2^{1/4}}\right)$$

$$= 3 + \frac{3}{2} \cdot \frac{\sqrt{2}}{2} = 3 + \sqrt{\frac{9}{8}}$$

$$\Rightarrow \left[\frac{9}{8}\right] + 3 = 1 + 3 = 4$$

75. (4)

The population after $2n$ days

$$= P\left(\left(\frac{3}{2} \cdot \frac{1}{2}\right) \cdot \left(\frac{3}{2} \cdot \frac{1}{2}\right) \cdot \ldots . n \text{ times}\right) = 3^{10}$$

$$\left(\frac{3}{4}\right)^n = \left(\frac{3}{4}\right)^{10} \Rightarrow n = 10$$

$\Rightarrow$ Total number of days = 20 days



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if